

Name	Umair Noor
Roll No	2500603
Submitted to	Sir Raja Faisal
Submitted By	Umair Noor

-

Lab 4 : Build a Student Grade Analyzer:

Task:

Create a Student class with name, marks in the constructor and a getGrade() method (A/B/C/F, same scale as Lecture 03).

- Create an array of at least 6 Student instances using new Student(...).
- Use filter() to get the passing students (marks >= 50) and map() to list just their names.
- Use reduce() to calculate the class average, and sort() + destructuring to find the top scorer's name.
- Use a ternary to build a one-line "Pass" / "Fail" label for each student as you display them.
- Log everything to the console with clear labels: passing names, class average, and the top scorer.

Code Screenshot:

```
1 // Student class
2 class Student {
3   constructor(name, marks) {
4     this.name = name;
5     this.marks = marks;
6   }
7
8   // Get grade according to marks
9   getGrade() {
10    if (this.marks >= 80) {
11      return "A";
12    } else if (this.marks >= 70) {
13      return "B";
14    } else if (this.marks >= 60) {
15      return "C";
16    } else {
17      return "F";
18    }
19  }
20 }
21
22 // Create an array of at least 6 Student objects
23 const students = [
24   new Student("Ali", 85),
25   new Student("Ahmed", 72),
26   new Student("Sara", 91),
27   new Student("Ayesha", 48),
28   new Student("Muhammad", 65),
29   new Student("Fatima", 55)
30 ];
31
32 // Display each student with Pass/Fail label using ternary
33 console.log("----- All Students -----");
34
35 students.forEach((student) => {
36   const label = student.marks >= 50 ? "Pass" : "Fail";
37   console.log(
38     `${student.name} - Marks: ${student.marks} - Grade: ${student.getGrade()} - ${label}`
39   );
40 });
41
42
43
44
45
```

```
24 const students = [
25   // ... (previous code) ...
31 ];
32
33 // Display each student with Pass/Fail label using ternary
34 console.log("----- All Students -----");
35
36 students.forEach((student) => {
37   const label = student.marks >= 50 ? "Pass" : "Fail";
38   console.log(
39     `${student.name} - Marks: ${student.marks} - Grade: ${student.getGrade()} - ${label}`
40   );
41 });
42
43 // filter() -> get passing students
44 const passingStudents = students.filter((student) => student.marks >= 50);
45
46 // map() -> get only passing students' names
47 const passingNames = passingStudents.map((student) => student.name);
48
49 // reduce() -> calculate class average
50 const totalMarks = students.reduce((total, student) => {
51   return total + student.marks;
52 }, 0);
53 const classAverage = totalMarks / students.length;
54
55 // sort() + destructuring -> find top scorer
56 const sortedStudents = [...students].sort((a, b) => b.marks - a.marks);
57 const [topScorer] = sortedStudents;
58
59 // Final results
60 console.log("\n----- Results -----");
61 console.log("Passing Names:", passingNames);
62 console.log("Class Average:", classAverage.toFixed(2));
63
64
65
66
67
68
69
70
71
72
73
```

Code Screenshot:

```
JS studentjs X
// const passingNames = passingStudents.map(student) => student.name;
52
53
54 // reduce() -> calculate class average
55 const totalMarks = students.reduce((total, student) => {
56   return total + student.marks;
57 }, 0);
58
59 const classAverage = totalMarks / students.length;
60
61
62 // sort() + destructuring -> find top scorer
63 const sortedStudents = [...students].sort((a, b) => b.marks - a.marks);
64
65 const [topScorer] = sortedStudents;
66
67
68 // Final results
69 console.log("\n----- Results -----");
70
71 console.log("Passing Names:", passingNames);
72
73 console.log("Class Average:", classAverage.toFixed(2));
74
75 console.log("Top Scorer:", topScorer.name);
76
```

Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Ahmed - Marks: 72 - Grade: B - Pass
Sara - Marks: 91 - Grade: A - Pass
Ayesha - Marks: 48 - Grade: F - Fail
Hamza - Marks: 65 - Grade: C - Pass
Fatima - Marks: 55 - Grade: F - Pass

----- Results -----
Passing Names: [ 'Ali', 'Ahmed', 'Sara', 'Hamza', 'Fatima' ]
Class Average: 69.33
Top Scorer: Sara
PS C:\Users\2500603\Desktop\Lab4>
```

***End ***